Practice Quiz 2 Solutions

1.1) {5, 4, 9, 2}

1.2) calculate() returns 18.

total is updated like so in the loop:

0 → 5 → 7 → 16 → 18

1.3) {4, 5, 8, 3}

1.4) Output:

```
18
4
8
```

1.5) No. result is assigned to the returned value of calculate(). change() does not affect result.

2.1)

6 4 6

3 10 12

6 7 14

2.2) 7

2.3) row == col selects the main diagonal.
row < col selects the cells above the main diagonal
else selects the cells below the main diagonal

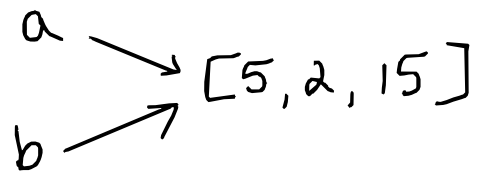
2.4 yes, with >6 instead of ≥6, count is 4 instead of 7.

3.1) output:

5
10
5
2
15

3.2) both a and b refer to the same origional array: {5, 2, 15}. The new array created inside mystery is not reachable through a or b in main.

Memory diagram:



3.3) Assigning y = 10 only changes the primitive variable y. Both a and b hold references to the same array, so b[2] = 15 changes the array seen through either reference

3.4) Java passes a copy of the reference value into parameter data. Reassigning data to a new array does not change the local variable a in main

4.1) one possible solution:

```
public static int countAvailable(char[] seats){
    int count = 0;

    for (int row = 0; row < seats.length;
                                i++) {
        for(int col = 0; col < seats[row].
                                length; col++) {
            if (seats[row][col] == '0') {
                count ++;

        }
    }
}
    return count;
}
```

4.2) one possible solution:

```
public static boolean hasAvailSeat(char [] raw) {

    for(int i =0; i< row.length; i++) {
        if (raw [i] == '0') {
            return true;
        }
    }

    return false;
}
```

4.3)

Iterate through each row, calling hasAvailSeat() on each. Return true as soon as one row contains an available seat. If none do, return false after the loop.

(code on next page)

4.3)

```
public static boolean seatAvail(char[][]
                                theatre){

    for(int row = 0;
        row < theatre.length;
        row++){
      if(hasAvailSeat(theatre[row]){
            return true;
      }
    }
    return false;
}
```